

Data Profiling Report: Patient Records (EHR)

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1.0 (Final)	2024-12-15	Layla Al-Fahad	Finalized for DGC submission

Approval

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1. EXECUTIVE SUMMARY

1.1 Overall Data Quality Score

The Epic EHR PATIENT_MASTER table was profiled using Informatica Data Quality (IDQ) v10.5 over a 5-day period (December 10-15, 2024). The table contains 18,500 active patient records across 45 columns, representing the Golden Record for all patients treated at Company A facilities.

Overall Data Quality Score: 96.2% (Excellent)

This score is calculated as a weighted average across six data quality dimensions:

Dimension	Score	Weight	Contribution	Status
Completeness	98.0%	25%	24.5%	■ Excellent
Accuracy	94.0%	25%	23.5%	■ Excellent
Consistency	97.5%	20%	19.5%	■ Excellent
Validity	95.8%	15%	14.4%	■ Excellent
Uniqueness	99.5%	10%	10.0%	■ Excellent

Timeliness	96.2%	5%	4.8%	■ Excellent
Weighted Overall		100%	96.2%	■ Excellent

1.2 Key Findings

Strengths

- High Uniqueness: 99.5% of patient records are unique with no duplicate National IDs (only 92 potential duplicates flagged for manual review)
- Excellent Completeness: 98.0% of critical fields are populated, significantly exceeding the 95% target for Critical Data Elements (CDEs)
- Strong Primary Key Integrity: PATIENT_UUID has 100% uniqueness and 0% NULL values
- Comprehensive Demographics: Core demographic fields (Name, Gender, Date of Birth) have >97% completeness
- Recent Updates: 96.2% of patient contact information has been updated within the past 12 months, demonstrating good timeliness

Weaknesses

- Inactive Mobile Numbers: 6% of mobile numbers (1,110 patients) are inactive based on SMS bounce detection - impacts ability to deliver critical lab results
- Missing National IDs: 2% of patient records (370 patients) lack National ID, primarily emergency admissions - creates billing and insurance verification issues
- Default Date Problem: 150 patient records have Date of Birth = 1900-01-01 (default value), indicating lazy data entry
- Invalid Email Formats: 4.2% of email addresses (780 patients) fail regex validation - impacts email communication effectiveness
- Address Standardization: 12% of addresses lack standardized formatting (e.g., "Riyadh" vs "■■■■■■■■" vs "RUH") - creates reporting inconsistencies

1.3 Critical Issues

The following critical data quality issues require immediate remediation:

Issue #1: Inactive Mobile Numbers (HIGH PRIORITY)

- Impact: Cannot deliver critical lab results via SMS to 1,110 patients (6% of patient base)
- Regulatory Risk: Healthcare regulations require notification of critical results within 24 hours
 - Patient Safety Risk: Patients may not receive time-sensitive health information

- Business Impact: Contributed to 8% no-show rate for appointments (180 missed appointments in Q4 2024)

Root Cause:

- Patients change mobile numbers without notifying the hospital
- No automated verification process at patient check-in
- Some patient records are 5+ years old with original mobile numbers

Recommended Action:

- Implement automated SMS verification at patient check-in (OTP code validation) - Target: Feb 15, 2025
- Launch "Update Your Contact Info" patient portal campaign - Target: Jan 30, 2025
- Implement automated SMS bounce detection and flagging - Target: Q1 2025

Issue #2: Missing National IDs (MEDIUM PRIORITY)

Impact: 370 patients (2%) lack National ID, creating operational challenges

- Billing Impact: Cannot query national health insurance database for coverage verification
- Integration Impact: Cannot integrate with national immunization registry
- Revenue Impact: Delays in billing submission for government-funded healthcare programs

Root Cause:

- Emergency admissions prioritize immediate care over administrative data collection
- National ID field not mandatory in EHR workflow (can be bypassed)
- Follow-up data collection process exists but not consistently executed

Recommended Action:

- Make National ID mandatory field for all patient registrations (emergency bypass allowed with required follow-up task)
- Retrospective outreach campaign to collect missing National IDs (call 370 patients) - Target: Feb 28, 2025
- Monthly reporting to Medical Services Director on incomplete patient records

Issue #3: Default Date Syndrome (MEDIUM PRIORITY)

Impact: 150 patient records have Date of Birth = 1900-01-01 (default value)

- Clinical Risk: Age-based clinical decision support alerts fail (e.g., pediatric medication dosing)
- Reporting Impact: Age demographics reported to government agencies are inaccurate
- Insurance Impact: Claims may be rejected due to impossible patient ages

Root Cause:

- Date picker fields default to `1900-01-01` if user doesn't actively select a date
- Users select default date to quickly bypass required field validation
- Training gap: Staff not trained on importance of accurate DOB for clinical safety

Recommended Action:

- Remove default date from Epic EHR date picker fields (force active selection) - Target: Q1 2025
- Implement date validation rules (DOB must be between 1920 and current date) - Target: Q1 2025

- Retrospective data collection: Contact 150 patients to collect actual DOB - Target: Q2 2025

2. TABLE SUMMARY

2.1 Structural Overview

Table Metadata

Attribute	Value
Table Name	PATIENT_MASTER
Schema	PRD_EHR
Database	Oracle 19c (Production)
Table Type	Heap-organized table
Row Count	18,500 active patients
Columns	45 columns
Indexes	8 indexes (1 primary key, 7 secondary)
Partitioning	None (small table size)
Compression	OLTP compression enabled
Table Size	85 GB (includes indexes and LOBs)

Column Distribution by Data Type

Data Type	Column Count	% of Total	Examples
VARCHAR2	28	62%	FIRST_NAME, LAST_NAME, ADDRESS_LINE1
NUMBER	8	18%	PATIENT_ID, AGE_YEARS, HEIGHT_CM
DATE	6	13%	DATE_OF_BIRTH, REGISTRATION_DATE, LAST_V
CHAR	2	4%	GENDER, MARITAL_STATUS
CLOB	1	2%	CLINICAL_NOTES (free-text)

Primary Key

Column	Data Type	Constraint	Uniqueness	Null Count
PATIENT_UUID	VARCHAR2(36)	PRIMARY KEY	100% (18,500 unique)	0 (0%)

Analysis: Primary key integrity is perfect - no NULL values, no duplicates. UUID format: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx

Foreign Keys

FK Name	Referencing Column	Referenced Table	Referential Integrity
FK_PHYSICIAN	PRIMARY_PHYSICIAN_ID	PHYSICIAN_MASTER	98.5% (orphaned: 1.5%)
FK_INSURANCE	INSURANCE_PROVIDER_ID	INSURANCE_PROVIDER	95.2% (orphaned: 4.8%)

Analysis: Minor referential integrity issues identified - 1.5% of patients reference non-existent physicians (likely retired physicians)

2.2 Data Volume & Growth

Historical Growth Trend

Period	Active Patients	New Registrations	Growth Rate
Q4 2023	16,200	420	+2.7%
Q1 2024	16,800	600	+3.7%
Q2 2024	17,400	600	+3.6%
Q3 2024	18,000	600	+3.4%
Q4 2024	18,500	500	+2.8%

Analysis: Consistent quarterly growth averaging +3.2% per quarter. Q4 2024 growth slightly slower (+2.8%) due to seasonal factors.

Patient Demographics Summary

Demographic	Count	% of Total
Total Active Patients	18,500	100%
Male	9,620	52%
Female	8,880	48%
Saudi Nationals	14,800	80%
Expatriates	3,700	20%
Age 0-18 (Pediatric)	4,625	25%
Age 18-60 (Adult)	11,100	60%

Age 60+ (Geriatric)	2,775	15%
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2.3 Data Freshness

Insert Activity

Metric	Value
Last Insert Timestamp	2024-12-15 09:30:00 (Today)
Inserts in Last 24 Hours	18 new patient registrations
Inserts in Last 7 Days	125 new patient registrations
Inserts in Last 30 Days	500 new patient registrations

Update Activity

Metric	Value
Last Update Timestamp	2024-12-15 10:15:00 (Today)
Updates in Last 24 Hours	342 records updated (1.8% of total)
Updates in Last 7 Days	2,405 records updated (13.0% of total)
Updates in Last 30 Days	8,880 records updated (48.0% of total)

Analysis: High update activity indicates active patient record maintenance. 48% of records updated in the past 30 days demonstrates good data currency.

Timeliness Analysis by Column

Column	Freshness Metric	Target	Actual	Status
PHONE_NUMBER	Updated within 12 months	90%	96.2%	■ Exceeds target
EMAIL_ADDRESS	Updated within 12 months	90%	94.8%	■ Exceeds target
ADDRESS_LINE1	Updated within 12 months	90%	92.5%	■ Exceeds target
INSURANCE_PROVIDER	Updated within 12 months	95%	98.2%	■ Exceeds target

3. COLUMN PROFILES (CRITICAL FIELDS)

3.1 PATIENT_UUID (Primary Key)

Column Metadata

Attribute	Value
Column Name	PATIENT_UUID
Data Type	VARCHAR2(36)
Nullable	NO (NOT NULL constraint)
Constraint	PRIMARY KEY
Index	PK_PATIENT_UUID (unique index)

Statistical Summary

Metric	Value	Status
Total Rows	18,500	N/A
Populated Rows	18,500 (100%)	■ Perfect
NULL Rows	0 (0%)	■ Perfect
Distinct Values	18,500 (100%)	■ Perfect
Duplicate Values	0 (0%)	■ Perfect

Pattern Analysis

UUID Format: xxxxxxxx-xxxx-4xxx-yxxx-xxxxxxxxxxxx (UUID v4 standard)

Sample Values:

- `a3d7f2c1-8b4e-4a1c-9f2d-3e8a7b6c5d4f`
- `7e9f1a2b-3c4d-4e5f-6a7b-8c9d0e1f2a3b`
- `2c4e6a8b-9d0f-4e1c-3a5b-7c9d1e3f5a7b`

Validation Result: 100% of UUIDs conform to UUID v4 standard format (verified using regex pattern matching)

Data Quality Assessment

Quality Dimension	Score	Findings
Completeness	100%	All rows have PATIENT_UUID populated
Uniqueness	100%	Zero duplicate UUIDs detected
Validity	100%	All UUIDs conform to UUID v4 standard format

Overall Column Score: 100% (Perfect)

3.2 NATIONAL_ID

Column Metadata

Attribute	Value
Column Name	NATIONAL_ID
Data Type	VARCHAR2(10)
Nullable	YES (should be NOT NULL for mandatory field)
Constraint	UNIQUE (duplicate check enforced)
Index	IDX_NATIONAL_ID (non-unique index for performance)
Business Rules	Saudi National ID: 10 digits, starts with 1 or 2<b

Statistical Summary

Metric	Value	Status
Total Rows	18,500	N/A
Populated Rows	18,130 (98.0%)	■■ Below 100% target
NULL Rows	370 (2.0%)	■■ Issue: Missing National IDs
Distinct Values	18,088 (99.8% of populated)	■ Good uniqueness
Duplicate Values	42 (0.2%)	■■ Potential duplicates flagged

Pattern Analysis

National ID Format Distribution:

Pattern	Description	Count	% of Total
1xxxxxxxxx	Saudi Male (starts with 1)	8,140	44.0%
2xxxxxxxxx	Saudi Female / Expatriate (starts with 2)	9,890	53.5%
Valid Total	Conforms to 10-digit pattern	18,030	97.5%
Invalid	Does not conform to expected pattern	100	0.5%
NULL	Missing National ID	370	2.0%

Breakdown of "Saudi Female vs Expatriate" (2xxxxxxxxx):

Using additional logic based on NATIONALITY field:

- Saudi Female: 7,400 records (40.0%)
- Expatriate (Iqama): 2,490 records (13.5%)

Invalid Pattern Examples

National ID	Count	Issue
0000000000	50	Default/placeholder value
9999999999	12	Default/placeholder value
12345	8	Too short (only 5 digits)
A123456789	5	Contains alphabetic characters
1234567890123	3	Too long (13 digits instead of 10)
3xxxxxxxxx	22	Invalid starting digit (should be 1 or 2)
Total Invalid	100	0.5% of total records

Duplicate Analysis

42 potential duplicate National IDs identified:

National ID	Occurrence Count	Patient Names	Analysis
1234567890	3	Ahmed Ali, Ahmed Ali Al-Rashid, Ahmed Ali Al-Rashid	Likely duplicates - same person registered multiple times
2345678901	2	Fatima Al-Zahrani, Fatima Zahrani	Likely duplicate - name variation
3456789012	2	Mohammad Al-Qahtani, M. Al-Qahtani	Likely duplicate - abbreviated name
...	...	...	...

Recommended Action: Manual review of 42 duplicate National IDs to determine if same person or data entry error

Data Quality Assessment

Quality Dimension	Score	Findings
Completeness	98.0%	370 missing National IDs (2.0%) - primarily emerge from recent registrations
Validity	99.5%	100 invalid formats (0.5%) - placeholder values and non-standard characters
Uniqueness	99.8%	42 duplicate National IDs (0.2%) - requires manual review
Consistency	100%	Cross-check with NATIONALITY field shows 100% logical consistency

Overall Column Score: 98.3% (Excellent, with minor issues)

3.3 FIRST_NAME & LAST_NAME

Column Metadata

Attribute	FIRST_NAME	LAST_NAME
Data Type	VARCHAR2(100)	VARCHAR2(100)
Nullable	NO (NOT NULL)	NO (NOT NULL)
Character Set	UTF-8 (supports Arabic and Latin characters)	UTF-8

Statistical Summary

Metric	FIRST_NAME	LAST_NAME
Populated Rows	18,500 (100%)	18,500 (100%)
NULL Rows	0 (0%)	0 (0%)
Distinct Values	2,840 unique first names	4,120 unique last names
Average Length	12.4 characters	18.6 characters
Min Length	2 characters (e.g., "Li")	3 characters (e.g., "Ali")
Max Length	45 characters	68 characters

Character Set Distribution

FIRST_NAME:

- Arabic script: 14,800 records (80%)
- Latin script: 3,200 records (17%)
- Mixed (Arabic + Latin): 500 records (3%)

LAST_NAME:

- Arabic script: 14,900 records (80.5%)
- Latin script: 3,100 records (16.8%)
- Mixed: 500 records (2.7%)

Common Values (Top 10)

FIRST_NAME:

Name	Count	% of Total
■■■■■ (Mohammad)	1,240	6.7%
■■■■■ (Ahmed)	980	5.3%

██████ (Fatima)	820	4.4%
████ (Ali)	780	4.2%
█████ (Sara)	680	3.7%

LAST_NAME:

Name	Count	% of Total
██ █████ (Al-Saud)	420	2.3%
██████████ (Al-Otaibi)	380	2.1%
███████████ (Al-Qahtani)	340	1.8%
███████████ (Al-Ghamdi)	320	1.7%
██████████ (Al-Harbi)	300	1.6%

Data Quality Issues

Issue #1: Leading/Trailing Spaces

- FIRST_NAME: 180 records (1.0%) have leading or trailing spaces
- LAST_NAME: 220 records (1.2%) have leading or trailing spaces
- Example: ` ` Ahmed ` ` instead of `Ahmed` `

Issue #2: Multiple Internal Spaces

- FIRST_NAME: 95 records (0.5%) have multiple consecutive spaces
- LAST_NAME: 140 records (0.8%) have multiple consecutive spaces
- Example: `Ahmed Ali` (double space) instead of `Ahmed Ali`

Issue #3: Mixed Case Inconsistency

- FIRST_NAME: 420 records (2.3%) have inconsistent capitalization
- Examples: `aHMED`, `AHMED` (all caps), `ahmed` (all lowercase)
- Expected: `Ahmed` (title case)

Data Quality Assessment

Quality Dimension	Score	Findings
Completeness	100%	All rows have FIRST_NAME and LAST_NAME populated
Validity	97.0%	3.0% have formatting issues (spaces, capitalizatio
Consistency	95.5%	4.5% have inconsistent character sets or capitaliz

Overall Column Score: 97.5% (Excellent)

Recommended Action: Implement automated TRIM() and title case standardization

3.4 DATE_OF_BIRTH

Column Metadata

Attribute	Value
Column Name	DATE_OF_BIRTH
Data Type	DATE
Nullable	NO (NOT NULL constraint)
Business Rules	Must be between 1920-01-01 and current date Age

Statistical Summary

Metric	Value	Status
Total Rows	18,500	N/A
Populated Rows	18,500 (100%)	■ Perfect
NULL Rows	0 (0%)	■ Perfect
Min Date	1900-01-01	■■ Default value detected
Max Date	2024-12-15	■ Newborns (valid)
Median Date	1985-06-12	■ Reasonable (39 years old)
Mean Date	1987-04-20	■ Reasonable (37 years old avg)

Age Distribution

Age Range	Count	% of Total	Category
0-1 years (Newborns)	420	2.3%	Pediatric
1-5 years (Toddlers)	1,280	6.9%	Pediatric
6-12 years (Children)	1,540	8.3%	Pediatric
13-17 years (Adolescents)	1,385	7.5%	Pediatric
18-60 years (Adults)	11,100	60.0%	Adult
61-80 years (Seniors)	2,405	13.0%	Geriatric
81+ years (Elderly)	370	2.0%	Geriatric
Total	18,500	100%	

Outliers & Anomalies

- Issue #1: Default Date (1900-01-01)
- Count: 150 records (0.8%)
 - Analysis: Indicates lazy data entry - users selected default date picker value
 - Impact: Patients shown as 124 years old in reports (2024 - 1900)
 - Sample Records:
 - PATIENT_UUID: `a3d7f2c1-8b4e-4a1c-9f2d-3e8a7b6c5d4f`, DOB: `1900-01-01`
 - PATIENT_UUID: `7e9f1a2b-3c4d-4e5f-6a7b-8c9d0e1f2a3b`, DOB: `1900-01-01`

- Issue #2: Future Dates (Born in the future)
- Count: 14 records (0.08%)
 - Analysis: Data entry error - likely typo in year (e.g., 2025 instead of 2024)
 - Sample Records:
 - PATIENT_UUID: `2c4e6a8b-9d0f-4e1c-3a5b-7c9d1e3f5a7b`, DOB: `2025-03-15` (3 months in the future)
 - PATIENT_UUID: `4f6a8c0e-1d3f-4b5a-7c9e-2d4f6a8b0c2d`, DOB: `2025-06-20` (6 months in the future)

- Issue #3: Impossibly Old Ages (>100 years)
- Count: 22 records (0.1%)
 - Analysis: Likely valid centenarians but should be verified
 - Oldest Patient: 106 years old (DOB: 1918-08-22)

Year Distribution (Top 10 Birth Years)

Birth Year	Count	% of Total	Age (as of 2024)
1985	920	5.0%	39 years
1990	880	4.8%	34 years
1988	820	4.4%	36 years
1992	780	4.2%	32 years
1987	750	4.1%	37 years
1995	720	3.9%	29 years
1982	680	3.7%	42 years
2010	640	3.5%	14 years
2015	580	3.1%	9 years
2020	520	2.8%	4 years

Data Quality Assessment

Quality Dimension	Score	Findings
Completeness	100%	All rows have DATE_OF_BIRTH populated
Validity	99.1%	164 invalid dates (0.9%) - future dates and default
Accuracy	99.2%	150 records with default date (0.8%) likely inaccu

Reasonableness	99.9%	22 centenarians (0.1%) should be verified but may
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Overall Column Score: 99.1% (Excellent, with minor issues)

Recommended Actions:

- 1. Remove default date from Epic EHR date picker (force active selection)
- 2. Implement validation rule: DOB must be between 1920-01-01 and current date
- 3. Contact 150 patients to collect actual DOB and update records
- 4. Verify 22 centenarians (age >100) with medical records staff

3.5 GENDER

Column Metadata

Attribute	Value
Column Name	GENDER
Data Type	CHAR(1)
Nullable	NO (NOT NULL constraint)
Allowed Values	'M' (Male), 'F' (Female)

Statistical Summary

Metric	Value	Status
Total Rows	18,500	N/A
Populated Rows	18,500 (100%)	■ Perfect
NULL Rows	0 (0%)	■ Perfect
Distinct Values	2 ('M', 'F')	■ Conforms to domain

Value Distribution

Value	Count	% of Total	Description
M (Male)	9,620	52.0%	Male patients
F (Female)	8,880	48.0%	Female patients
Total	18,500	100%	

Analysis: Gender distribution is balanced (52% Male, 48% Female), which is consistent with general population demographics.

Data Quality Assessment

Quality Dimension	Score	Findings
Completeness	100%	All rows have GENDER populated
Validity	100%	All values conform to allowed domain ('M' or 'F')
Consistency	See Cross-Column Analysis	Cross-check with TITLE field for logical consisten

Overall Column Score: 100% (Perfect)

3.6 PHONE_NUMBER (Mobile)

Column Metadata

Attribute	Value
Column Name	PHONE_NUMBER
Data Type	VARCHAR2(15)
Nullable	YES (optional field)
Business Rules	Saudi mobile format: +966 5XX XXX XXX Internati

Statistical Summary

Metric	Value	Status
Total Rows	18,500	N/A
Populated Rows	17,020 (92.0%)	■ Good completeness
NULL Rows	1,480 (8.0%)	■■ Missing phone numbers
Distinct Values	16,880 (99.2% of populated)	■ Good uniqueness

Format Distribution

Format	Example	Count	% of Populated	Status
+966 5XX XXX XXX	+966 501 234 567	14,468	85.0%	■ Correct international format
05XX XXX XXX	0501 234 567	1,702	10.0%	■■ Legacy local format
00966 5XX XXX XXX	00966 501 234 567	680	4.0%	■■ Alternative international format
5XX XXX XXX	501 234 567	170	1.0%	■■ Missing country code

Total Populated		17,020	100%	
NULL		1,480	N/A	■■ Missing phone numbers

Accuracy Validation (SMS Bounce Detection)

In Q4 2024, Company A implemented automated SMS bounce detection to validate mobile number accuracy:

Status	Count	% of Populated	Description
Active	15,910	93.5%	SMS successfully delivered
Inactive/Bounced	1,110	6.5%	SMS failed to deliver (disconnected numbers)

Critical Finding: 1,110 mobile numbers (6.5% of populated) are inactive - patients will not receive critical health notifications

Data Quality Issues

Issue #1: Inconsistent Formatting

- 2,552 records (15% of populated) use non-standard formats
- Creates challenges for SMS campaigns and call center dialer integration
- Requires standardization to +966 5XX XXX XXX format

Issue #2: Inactive Mobile Numbers

- 1,110 records (6.5% of populated) have inactive/disconnected numbers
- High-priority issue affecting patient safety (cannot deliver critical lab results)

Issue #3: Duplicate Phone Numbers

- 140 phone numbers shared by multiple patients (280 total records)
- May indicate family members or data entry errors

Data Quality Assessment

Quality Dimension	Score	Findings
Completeness	92.0%	1,480 missing phone numbers (8.0%)
Validity	85.0%	15.0% use non-standard formats (need standardizati
Accuracy	93.5%	6.5% inactive numbers (SMS bounce detection)
Uniqueness	99.2%	0.8% duplicate phone numbers (family members or er

Overall Column Score: 92.0% (Good, with notable accuracy issues)

Recommended Actions:

1. Implement automated phone number standardization (convert all to +966 5XX XXX XXX)
2. Implement SMS verification at patient check-in (OTP code validation)

- 3. Launch "Update Your Contact Info" campaign via patient portal
- 4. Implement ongoing SMS bounce detection and flagging

3.7 EMAIL_ADDRESS

Column Metadata

Attribute	Value
Column Name	EMAIL_ADDRESS
Data Type	VARCHAR2(100)
Nullable	YES (optional field)
Business Rules	Must conform to RFC 5322 email format Must be d

Statistical Summary

Metric	Value	Status
Total Rows	18,500	N/A
Populated Rows	16,650 (90.0%)	■ Good completeness
NULL Rows	1,850 (10.0%)	■■ Missing email addresses
Distinct Values	16,520 (99.2% of populated)	■ Good uniqueness

Format Validation (Regex Pattern Matching)

Regex Pattern: ^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}\$

Status	Count	% of Populated	Examples
Valid Format	15,870	95.3%	ahmed.ali@gmail.com, fatima@outlook.sa
Invalid Format	780	4.7%	user@gmail..com, test@@domain.com, email@dom

Common Invalid Formats

Invalid Pattern	Count	Example
Double dots (..)	320	user@gmail..com
Double @ symbols	180	test@@domain.com
Missing domain extension	140	email@domain

Leading/trailing spaces	95	" user@email.com "
Special characters	45	user#name@email.com

Domain Distribution (Top 10)

Email Domain	Count	% of Populated
@gmail.com	6,240	37.5%
@outlook.com	2,820	16.9%
@hotmail.com	1,680	10.1%
@yahoo.com	1,250	7.5%
@outlook.sa	980	5.9%
@live.com	820	4.9%
@icloud.com	680	4.1%
@protonmail.com	420	2.5%
@companya.gov.sa (internal)	180	1.1%
Other domains	1,580	9.5%

Deliverability Validation

Note: Email deliverability validation not yet implemented (planned for Q1 2025 using SMTP validation API)

Data Quality Assessment

Quality Dimension	Score	Findings
Completeness	90.0%	1,850 missing email addresses (10.0%)
Validity	95.3%	780 invalid email formats (4.7%)
Uniqueness	99.2%	130 duplicate email addresses (0.8%)

Overall Column Score: 94.5% (Excellent)

Recommended Actions:

- 1. Implement real-time email validation on Epic EHR forms (reject invalid formats)
- 2. Deploy email verification service (SMTP validation) to validate existing database
- 3. Clean up 780 invalid email addresses (contact patients for correction)

3.8 ADDRESS_LINE1

Column Metadata

Attribute	Value
Column Name	ADDRESS_LINE1
Data Type	VARCHAR2(200)
Nullable	YES

Statistical Summary

Metric	Value	Status
Populated Rows	17,575 (95.0%)	■ Excellent completeness
NULL Rows	925 (5.0%)	■■ Minor gap
Average Length	42.8 characters	N/A
Distinct Values	14,820 (84.3% of populated)	■ Good diversity

Data Quality Issues

Issue #1: Inconsistent Language

- Arabic script: 60%
- Latin script: 35%
- Mixed (Arabic + Latin): 5%

Issue #2: Non-Standardized Abbreviations

- "St." vs "Street" vs "■■■■"
- "Dist." vs "District" vs "■■"

Overall Column Score: 92.5% (Good)

3.9 CITY & PROVINCE

Statistical Summary

Field	CITY	PROVINCE
Populated	96.5%	97.0%
Top Value	Riyadh (40%)	Riyadh Region (42%)

Overall Score: 95.0% (Excellent)

3.10 NATIONALITY

Value Distribution

Nationality	Count	% of Total
Saudi	14,800	80.0%
Egyptian	920	5.0%
Indian	680	3.7%
Pakistani	580	3.1%
Filipino	420	2.3%
Other	1,100	5.9%

Overall Score: 99.5% (Excellent)

3.11 MARITAL_STATUS

Value Distribution

Status	Count	% of Total
M (Married)	10,360	56.0%
S (Single)	6,660	36.0%
D (Divorced)	920	5.0%
W (Widowed)	560	3.0%

Overall Score: 100% (Perfect)

3.12 BLOOD_TYPE

Value Distribution

Blood Type	Count	% of Populated
O+	5,180	30.6%
A+	4,420	26.1%
B+	3,220	19.0%
AB+	1,680	9.9%

O-	1,020	6.0%
A-	780	4.6%
B-	520	3.1%
AB-	120	0.7%
Populated Total	16,940	100%
NULL	1,560	8.4% of total

Overall Score: 91.6% (Good)

4. CROSS-COLUMN DEPENDENCIES

4.1 Gender vs Title

Business Rule

Rule: If TITLE = 'Mr', then GENDER must = 'M'
Rule: If TITLE = 'Mrs' or 'Ms', then GENDER must = 'F'

Validation Results

Violation Type	Count	% of Total	Example Records
TITLE='Mr' but GENDER='F'	15	0.08%	PATIENT_UUID: a3d7f2c1...
TITLE='Mrs' but GENDER='M'	8	0.04%	PATIENT_UUID: 7e9f1a2b...
Total Violations	23	0.12%	

Analysis: Very low violation rate (0.12%) suggests data entry errors or transgender patients

Recommended Action: Manual review of 23 violating records with medical staff

4.2 Age vs Employment Status

Business Rule

Rule: If EMPLOYMENT_STATUS = 'Employed', then AGE must be >= 18

Validation Results

Status	Count	% of Total
Compliant	11,100	100%
Violations	0	0%

Analysis: 100% compliance - no minors (<18 years) marked as employed

4.3 Nationality vs National ID Format

Business Rule

Rule: If NATIONALITY = 'Saudi', then NATIONAL_ID should start with '1' or '2'

Rule: If NATIONALITY != 'Saudi', then NATIONAL_ID should start with '2' (Iqama)

Validation Results

Validation	Count	% of Total	Status
Logical Consistency	18,130	100%	■ Perfect
Violations	0	0%	■ Perfect

Analysis: 100% logical consistency between NATIONALITY and NATIONAL_ID format

5. DATA QUALITY RULES VALIDATION

5.1 Completeness Rules

Rule ID	Rule Description	Target	Actual	Status
COMP-001	PATIENT_UUID must not be NULL	100%	100%	■ Pass
COMP-002	NATIONAL_ID must not be NULL	100%	98.0%	■■ Fail (370 NULLs)
COMP-003	FIRST_NAME must not be NULL	100%	100%	■ Pass
COMP-004	LAST_NAME must not be NULL	100%	100%	■ Pass
COMP-005	DATE_OF_BIRTH must not be NULL	100%	100%	■ Pass
COMP-006	GENDER must not be NULL	100%	100%	■ Pass
COMP-007	PHONE_NUMBER completeness	95%	92.0%	■■ Fail (below target)

COMP-008	EMAIL_ADDRESS completeness	90%	90.0%	■ Pass (meets target)
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Overall Completeness Score: 98.0%

5.2 Validity Rules

Rule ID	Rule Description	Target	Actual	Status
VAL-001	NATIONAL_ID must be 10 digits	100%	99.5%	■■ Fail (100 invalid)
VAL-002	PHONE_NUMBER must match +8885XX format	100%	85.0%	■■ Fail (15% non-standard)
VAL-003	EMAIL_ADDRESS must match RFC322 format	100%	95.3%	■■ Fail (780 invalid)
VAL-004	DATE_OF_BIRTH must be between 1920 and today	100%	99.1%	■■ Fail (164 invalid)
VAL-005	GENDER must be 'M' or 'F'	100%	100%	■ Pass
VAL-006	BLOOD_TYPE must be valid ABO type	100%	100%	■ Pass

Overall Validity Score: 95.8%

5.3 Uniqueness Rules

Rule ID	Rule Description	Target	Actual	Status
UNQ-001	PATIENT_UUID must be unique	100%	100%	■ Pass
UNQ-002	NATIONAL_ID must be unique	100%	99.8%	■■ Fail (42 duplicates)
UNQ-003	EMAIL_ADDRESS uniqueness	99%	99.2%	■ Pass
UNQ-004	PHONE_NUMBER uniqueness	99%	99.2%	■ Pass

Overall Uniqueness Score: 99.5%

5.4 Consistency Rules

Rule ID	Rule Description	Target	Actual	Status
CON-001	GENDER vs TITLE consistency	100%	99.9%	■■ Fail (23 violations)
CON-002	NATIONALITY vs NATIONAL_ID format	100%	100%	■ Pass
CON-003	AGE vs EMPLOYMENT_STATUS	100%	100%	■ Pass

Overall Consistency Score: 99.9%

6. PATTERN ANALYSIS

6.1 National ID Patterns

Most Common Patterns:

- `1xxxxxxxx` (Saudi Male): 8,140 records (44%)
- `2xxxxxxxx` (Saudi Female/Expat): 9,890 records (53.5%)

Anomalous Patterns:

- `0000000000` (placeholder): 50 records
- `9999999999` (placeholder): 12 records
- Starting with `3-9` (invalid): 22 records

6.2 Phone Number Patterns

Standard Format: +966 5XX XXX XXX

Observed Formats:

- International (+966): 85%
- Local (05XX): 10%
- Alternative (00966): 4%
- Incomplete (5XX): 1%

6.3 Email Address Patterns

Valid Format: 95.3%

Invalid Formats:

- Double dots (..): 320 records
- Double @ (@@): 180 records
- Missing extension: 140 records

7. OUTLIERS & ANOMALIES

7.1 Default Values

Column	Default Value	Count	% of Total
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DATE_OF_BIRTH	1900-01-01	150	0.8%
NATIONAL_ID	0000000000	50	0.3%
NATIONAL_ID	9999999999	12	0.06%

Total Default Values: 212 records (1.1%)

7.2 Out-of-Range Values

Column	Issue	Count
DATE_OF_BIRTH	Future dates	14
AGE	>100 years	22

7.3 Suspicious Patterns

- Duplicate National IDs: 42 instances
- Same phone number for multiple patients: 140 numbers (280 records)

8. RECOMMENDATIONS

8.1 Critical Priority (Immediate Action)

1. Implement SMS Verification at Patient Check-in (Target: Feb 15, 2025)
 - Impact: Validates 1,110 inactive mobile numbers
 - Owner: IT Operations Manager
2. Remove Default Date from Epic EHR (Target: Q1 2025)
 - Impact: Prevents future 1900-01-01 entries
 - Owner: IT Development Lead
3. Retrospective National ID Collection (Target: Feb 28, 2025)
 - Impact: Collects 370 missing National IDs
 - Owner: Patient Registration Manager

8.2 High Priority (Q1 2025)

1. Implement Real-Time Email Validation

- 2. Automated Phone Number Standardization
- 3. Address Standardization Project

8.3 Medium Priority (Q2 2025)

- 1. Manual Review of Duplicate National IDs (42 records)
- 2. Verification of Centenarians (22 patients >100 years old)
- 3. Gender vs Title Violation Review (23 records)

9. APPENDIX

9.1 Full Column List (45 Columns)

#	Column Name	Data Type	Nullable	Description
1	PATIENT_UUID	VARCHAR2(36)	NO	Primary key (UUID v4)
2	PATIENT_ID	NUMBER	NO	Legacy numeric ID
3	NATIONAL_ID	VARCHAR2(10)	YES	Saudi National ID or Iqama
4	FIRST_NAME	VARCHAR2(100)	NO	Patient first name
5	LAST_NAME	VARCHAR2(100)	NO	Patient last name
6	MIDDLE_NAME	VARCHAR2(100)	YES	Patient middle name
7	TITLE	VARCHAR2(10)	YES	Mr, Mrs, Ms, Dr
8	DATE_OF_BIRTH	DATE	NO	Patient date of birth
9	AGE_YEARS	NUMBER	YES	Calculated age in years
10	GENDER	CHAR(1)	NO	M or F
11	NATIONALITY	VARCHAR2(50)	YES	Patient nationality
12	MARITAL_STATUS	CHAR(1)	YES	M, S, D, W
13	BLOOD_TYPE	VARCHAR2(5)	YES	ABO and Rh blood type
14	HEIGHT_CM	NUMBER	YES	Height in centimeters
15	WEIGHT_KG	NUMBER	YES	Weight in kilograms
16	PHONE_NUMBER	VARCHAR2(15)	YES	Mobile phone number
17	EMAIL_ADDRESS	VARCHAR2(100)	YES	Email address
18	ADDRESS_LINE1	VARCHAR2(200)	YES	Street address
19	ADDRESS_LINE2	VARCHAR2(200)	YES	Apartment/unit number
20	CITY	VARCHAR2(100)	YES	City name

21	PROVINCE	VARCHAR2(100)	YES	Province/region
22	POSTAL_CODE	VARCHAR2(10)	YES	Postal code
23	COUNTRY	VARCHAR2(50)	YES	Country (default: Saudi Arabia)
24	PRIMARY_PHYSICIAN_ID	VARCHAR2(36)	YES	FK to PHYSICIAN_MASTER
25	INSURANCE_PROVIDER_ID	VARCHAR2(36)	YES	FK to INSURANCE_PROVIDER
26	INSURANCE_POLICY_NO	VARCHAR2(50)	YES	Insurance policy number
27	EMERGENCY_CONTACT_NAME	VARCHAR2(200)	YES	Emergency contact full name
28	EMERGENCY_CONTACT_PHONE	VARCHAR2(15)	YES	Emergency contact phone
29	EMPLOYMENT_STATUS	VARCHAR2(20)	YES	Employed, Unemployed, Retired
30	OCCUPATION	VARCHAR2(100)	YES	Job title/occupation
31	EMPLOYER_NAME	VARCHAR2(200)	YES	Employer organization name
32	PREFERRED_LANGUAGE	VARCHAR2(20)	YES	AR (Arabic) or EN (English)
33	RELIGION	VARCHAR2(50)	YES	Patient religion
34	ALLERGY_FLAG	CHAR(1)	YES	Y or N
35	CHRONIC_DISEASE_FLAG	CHAR(1)	YES	Y or N
36	SMOKING_STATUS	VARCHAR2(20)	YES	Current, Former, Never
37	ALCOHOL_CONSUMPTION	VARCHAR2(20)	YES	Never (Islamic country)
38	REGISTRATION_DATE	DATE	NO	First registration date
39	REGISTRATION_SOURCE	VARCHAR2(50)	YES	Walk-in, Referral, Emergency
40	LAST_VISIT_DATE	DATE	YES	Most recent visit date
41	VISIT_COUNT	NUMBER	YES	Total visit count
42	STATUS	VARCHAR2(20)	NO	Active, Inactive, Deceased
43	CLINICAL_NOTES	CLOB	YES	Free-text clinical notes
44	CREATED_DATE	DATE	NO	Record creation timestamp
45	LAST_MODIFIED_DATE	DATE	NO	Record last modified timestamp

9.2 SQL Profiling Queries

Completeness Check

```

sql
SELECT
COLUMN_NAME,
COUNT() AS TOTAL_ROWS,
COUNT(COLUMN_NAME) AS POPULATED_ROWS,
COUNT() - COUNT(COLUMN_NAME) AS NULL_ROWS,
ROUND((COUNT(COLUMN_NAME) / COUNT()) 100, 2) AS COMPLETENESS_PCT

```

FROM PATIENT_MASTER
GROUP BY COLUMN_NAME;

Pattern Analysis (National ID)

sql
SELECT
CASE
WHEN NATIONAL_ID LIKE '1%' THEN '1xxxxxxxx (Saudi Male)'
WHEN NATIONAL_ID LIKE '2%' THEN '2xxxxxxxx (Saudi Female/Expat)'
WHEN LENGTH(NATIONAL_ID) != 10 THEN 'Invalid Length'
WHEN NATIONAL_ID NOT LIKE '[0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9]' THEN 'Non-Numeric'
ELSE 'Other Pattern'
END AS PATTERN,
COUNT() AS COUNT,
ROUND((COUNT() / (SELECT COUNT() FROM PATIENT_MASTER WHERE NATIONAL_ID IS NOT NULL))
100, 2) AS PCT
FROM PATIENT_MASTER
WHERE NATIONAL_ID IS NOT NULL
GROUP BY CASE...;

9.3 Glossary of Terms

Term	Definition
Completeness	The extent to which required data is present (no m
Validity	The extent to which data conforms to defined forma
Uniqueness	The extent to which records are unique (no duplica
Accuracy	The extent to which data correctly represents real
Consistency	The extent to which data is consistent across rela
Timeliness	The extent to which data is current and up-to-date
National ID	10-digit Saudi National ID or Iqama (expatriate ID
UUID	Universally Unique Identifier (128-bit value)
SMS Bounce	Failed SMS delivery indicating inactive mobile num

END OF REPORT
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